CHAPTER 9

# Matrices

Operations, inverse, powers, systems and transformations of the plane

**Objectives.** Compute with matrices (sum, product by a real number, product); find the inverse of a matrix of order 2, and of order 3 with a calculator; compute a power, in particular by induction; solve a linear system in matrix form; represent a transformation of the plane by a matrix.

**Prerequisites.** Linear systems of two equations; coordinates of vectors; addition formulas (chapter 3); proof by induction; sequences.

## Fluency

**Fluency · No calculator, five minutes.**

1. Solve the system $3x + 5y = 4$ and $x + 2y = 1$.
2. Compute the determinant of the vectors $\overrightarrow{a}\begin{pmatrix}3\\1\end{pmatrix}$ and $\overrightarrow{b}\begin{pmatrix}5\\2\end{pmatrix}$.
3. Give $\cos\frac{\pi}{3}$ and $\sin\frac{\pi}{3}$.
4. What are the coordinates of the image of the point $(x, y)$ under the reflection in the axis $(Ox)$?
5. Expand $(a+b)^2$. Do we use $ab = ba$?
6. Let $u_{n+1} = 2u_n$ and $u_0 = 3$. Express $u_n$ in terms of $n$.